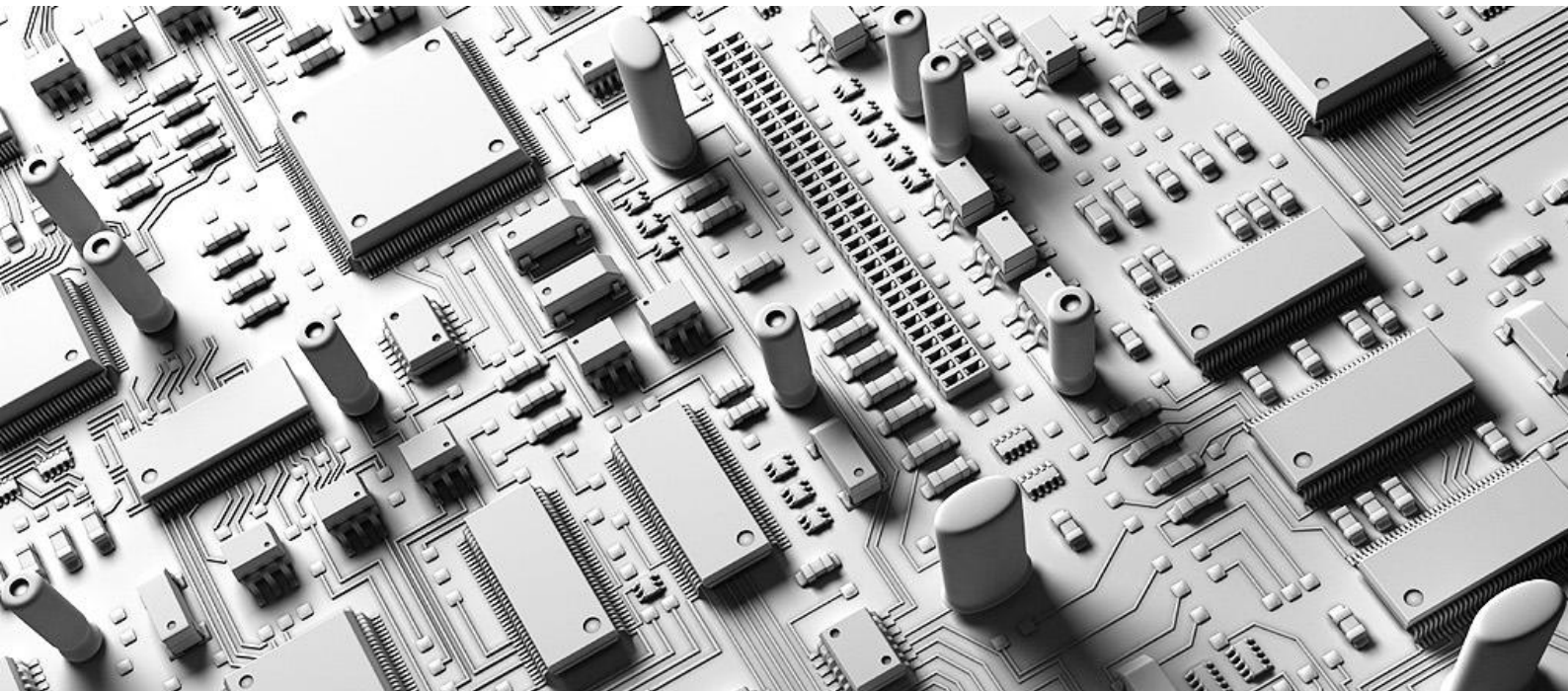




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G520 Mediatek AIoT Module

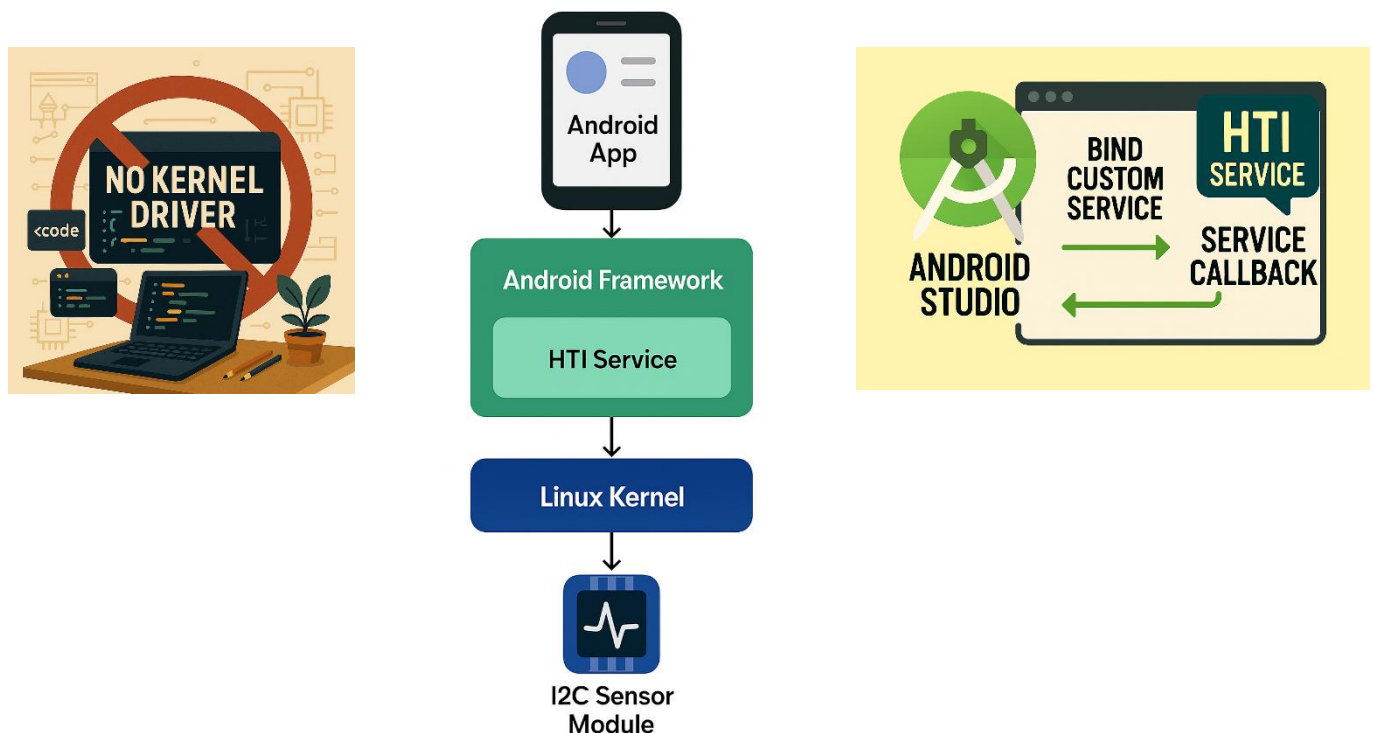
Powerful AI computing for AIoT endpoints, enabling rapid product development in diverse applications.

Overview

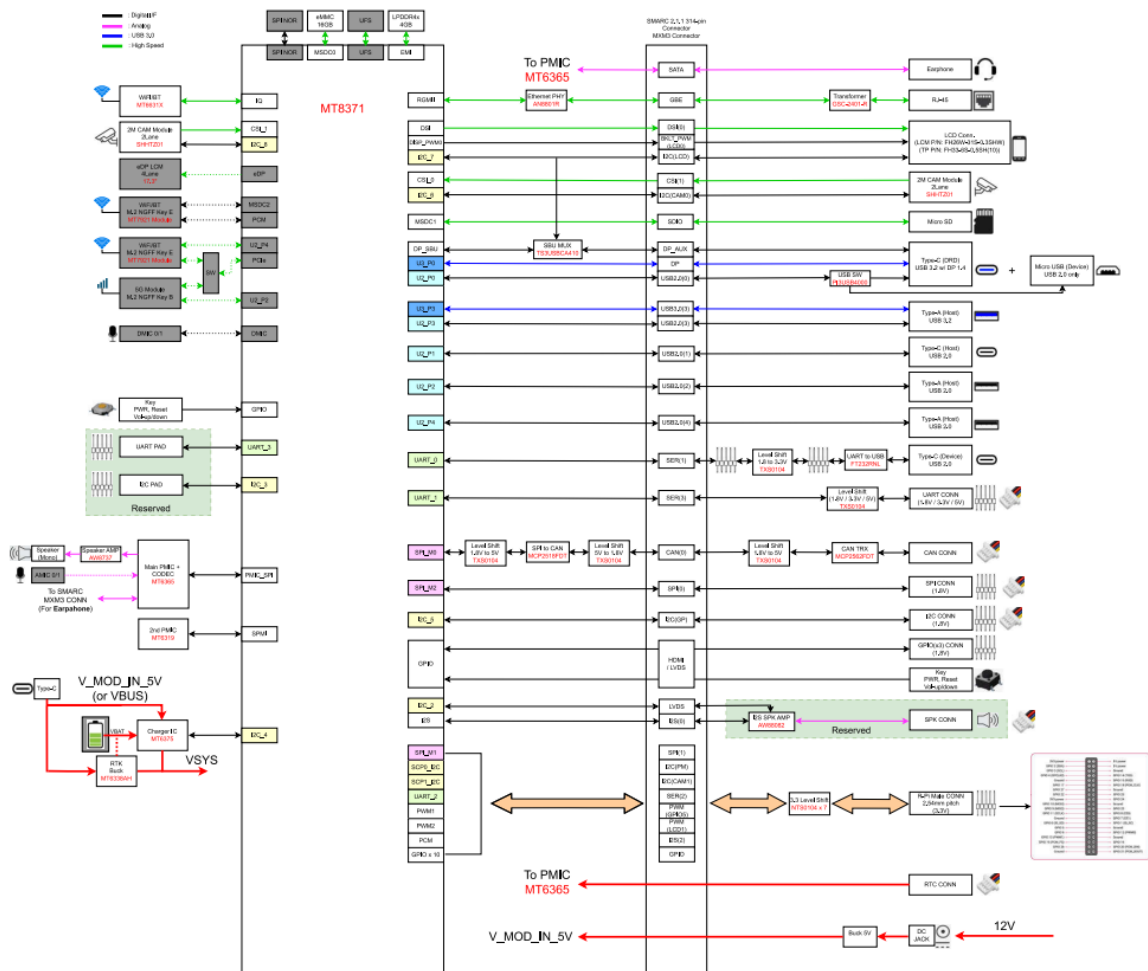
1 Introduction

The MediaTek Genio 520 platform features an octa-core ARM Cortex architecture paired with MediaTek's 8th Generation NPU, delivering up to 10 TOPS of AI computing power. This significantly enhances edge AI capabilities for generative AI applications, making it ideal for Industrial HMI, Smart Home, Retail and Transportation scenarios.

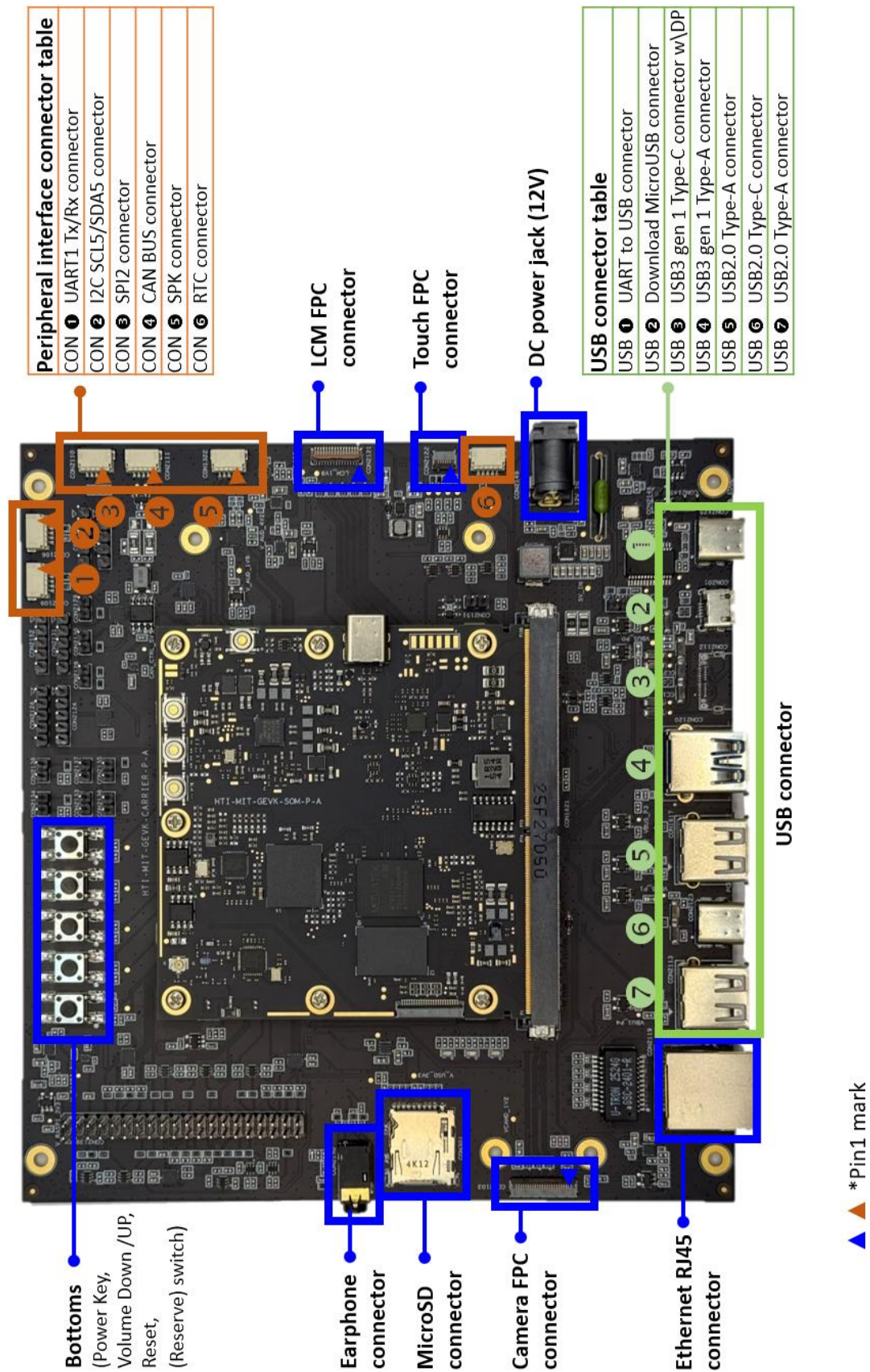
The platform integrates versatile interfaces, including MIPI-DSI, I2C, UART, SPI and several USB ports which supporting 2x USB 3.2 Gen1 and 3x USB 2.0. These USB interfaces enable easy connection to USB cameras, and diverse sensors, expanding entertainment system capabilities and IoT product functions. In order easier for customer to apply the module to implement their product, HTI offers an exclusive service designed specifically for Android-based sensor integration, you may access APIs during Android Studio development, it means you don't need setup build environment and porting kernel driver any more, HTI exclusive service support I2C, SPI, UART, CAN communication SDK, you can accelerate and simplify the development process for integrating external hardware modules.



2 Block Diagram



3 Peripheral I/O



4 PIN Definition

ZIF Connector PIN Definition

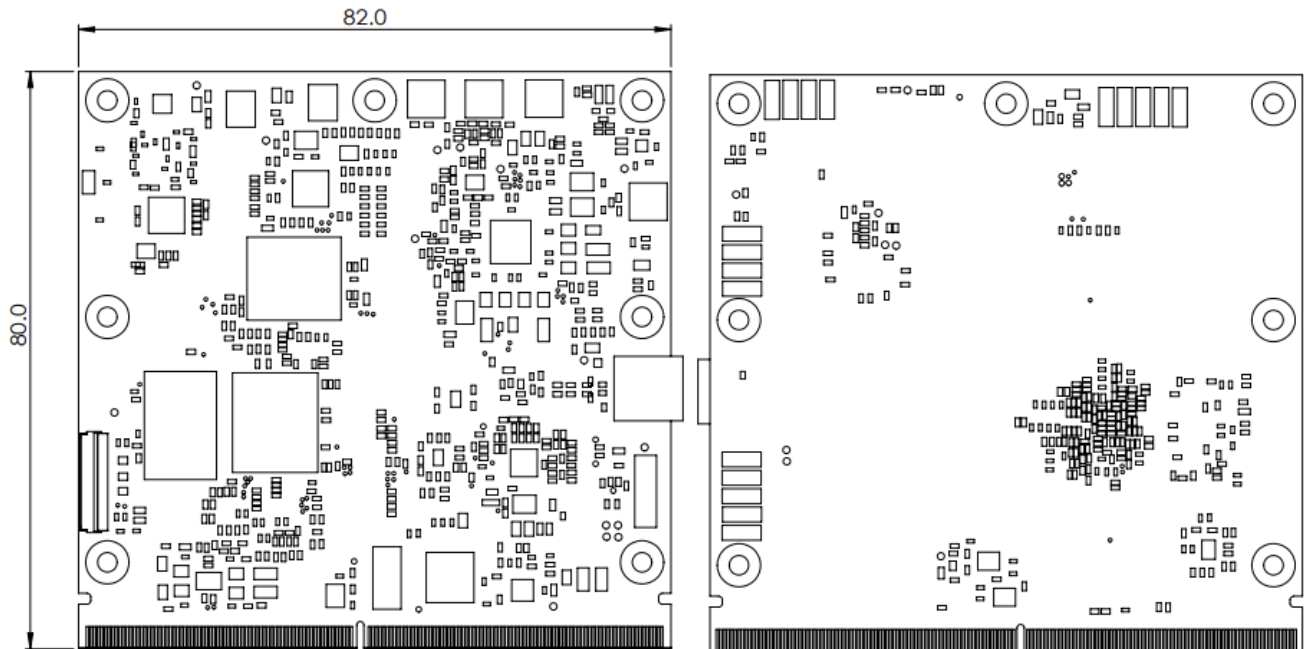
ZIF	Camera FPC connector	ZIF	LCM FPC connector
Pin	Signal Name	Pin	Signal Name
1	NC	1	VLEDA
2	DOVDD	2	VLEDA
3	AVDD	3	VLEDA
4	AVDD	4	NC
5	RESET	5	VLEDK
6	SCL	6	VLEDK
7	SDA	7	VLEDK
8	GND	8	VLEDK
9	MCLK	9	GND
10	GND	10	GND
11	MCP	11	MIPI_D2P
12	MCN	12	MIPI_D2N
13	GND	13	GND
14	MDP	14	MIPI_D1P
15	MDN	15	MIPI_D1N
16	GND	16	GND
17	MDP1	17	MIPI_CLKP
18	MDN1	18	MIPI_CLKN
19	GND	19	GND
20	DVDD	20	MIPI_D0P
21	DVDD	21	MIPI_D0N
22	GND	22	GND
23	GND	23	MIPI_D3P
24	GND	24	MIPI_D3N
ZIF	Touch FPC connector	25	GND
Pin	Signal Name	26	GND
1	SCL	27	RESET
2	SDA	28	VDD1V8
3	GND	29	VDD1V8
4	RST	30	VDD3V3
5	INT	31	VDD3V3
6	VDD33		

Peripheral Interface PIN Definition

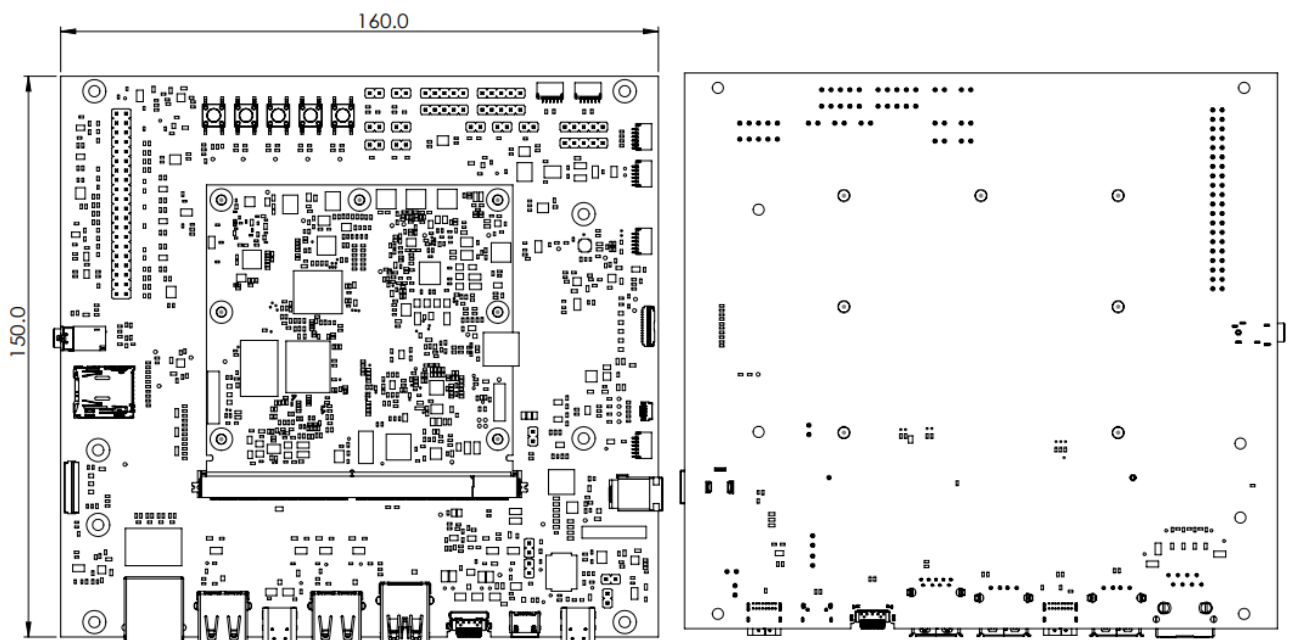
CON ❶	UART1 Tx/Rx connector	CON ❷	CAN BUS connector
Pin	Signal Name	Pin	Signal Name
1	NC	1	NC
2	UART1_RXD_LV	2	GND
3	NC	3	CAN_L
4	UART1_TXD_LV	4	CAN_H
5	GND	5	GND
CON ❸	I2C SCL5/SDA5 connector (1V8 Power domain)	CON ❹	SPK connector
Pin	Signal Name	Pin	Signal Name
1	NC	1	SPKR_P
2	SDA5	2	SPKR_N
3	NC	3	GND
4	SCL5	4	SPKL_P
5	GND	5	SPKL_N
CON ❹	SPI2 connector (1V8 Power domain)	CON ❺	RTC connector
Pin	Signal Name	Pin	Signal Name
1	SPIM2_MISO	1	NC
2	SPIM2_MOSI	2	NC
3	SPIM2_CS_N	3	NC
4	SPIM2_CLK	4	VRTC
5	GND	5	GND

5 Dimensions

SoM Board



Carrier Board



Module Spec

■ G520 SoM SPEC :

Processor	
SoC Platform	Genio 520 (MT8371+MT6365+MT6319+RT6375+MT6631X)
Memory/Storage	
RAM	4GB LPDDR4x (discrete)
eMMC	16GB TLC (discrete)
Wireless Connectivity	
Wi-Fi	WiFi-5 , IEEE 802.11 ac/a/b/g/n
Bluetooth	Bluetooth 5.3

■ G520 Carrier Board SPEC :

Onboard I/O	
SMARC Modules Socket	Compatibility SMARC 2.1
Peripheral I/O	
Network	GbE-RJ45 Jack
Display/TP	4-lane MIPI-DSI, Zif connector
DP	USBC DP-1.4
Camera	2-lane MIPI-CSI, Zif connector
Audio	3.5mm 4-pole Headset
USB	1 x DRD USB3.2 Type-C (w/ DP and download feature, USB-P0) 1 x Host USB 3.2 Type-A (USB-P3) 1 x Host USB2.0 Type-C 2 x Host USB2.0 Type-A
Serial Port	1.8V / 3.3V / 5V UART interface connector
CAN	CAN interface connector
SPI	1.8V / 3.3V SPI interfaces connector
I2C	1.8V / 3.3V I2C interfaces connector

Storage	microSD slot
PWM	3.3V PWM
GPIO	10 pins
Buttons	Power/Reset/Download/V+/V-/GPIO-x Button
Power	12V DC-in (2.1mm Inner Pin, 5.5mm Outer DC female)

■ G520 Software features :

Software	
Operating System	Android 14 (support to Android 15)
Integrated Development Environment (IDE)	Android Studio development
SDK	HTI service (Built-in HTI service; enables programming for I2C, UART, SPI, and more without the need to develop kernel drivers)
Firmware Update	Support update.zip by HTI
Root Access	Supported